Bayes Estimation

A Crash Course from First Principles

Parametric Inference, B.Stat. 3rd Year — Lectures 6–7

This assumes you know nothing about Bayes methods and everything about the first half of the course (risk, unbiasedness, sufficiency, UMVUE). Read it front to back — each section is built on the one before. Worked examples use real numbers so you can check every step.

Contents

1 Why Bayes exists at all

Go back to the very first thing proved in Lecture 1.

The problem the whole course is trying to solve

For an estimator T of θ we measure quality by the *risk*

$$R(\theta, T) = \mathbb{E}_\theta\left[(T-\theta)^2\right].$$

We would love to find the T that minimises this. **But we can't**, because $R(\theta, T)$ is a *function of* θ, not a number, and θ is unknown.
Concretely: whatever T you propose, I beat you at the single point θ_0 by proposing the silly estimator $T' \equiv \theta_0$, which has $R(\theta_0, T') = 0$. So no estimator is best everywhere, and "minimise the risk" is not a well-posed problem.

There are exactly three ways out of this, and the course does all three:

Escape route	What you do to $R(\theta, T)$	What you get
Restrict the class	Only compare estimators that are unbiased — this rules out the silly $T \equiv \theta_0$	**UMVUE** (Lectures 1–5)
Average it	Collapse the function $R(\cdot, T)$ to a single number by averaging over θ	**Bayes** (Lectures 6–7)
Take the worst case	Collapse it to a single number by taking $\sup_\theta$	**Minimax** (Lectures 8–9)

So what *is* Bayes estimation, in one sentence?

It is what you get when you turn the risk *function* into a single *number* by averaging it with weights, and then minimise that number.
The weights are called the **prior**. That is the whole idea. Everything else — posteriors, conjugacy, Beta and Gamma distributions — is machinery for actually doing the computation.

2 The one conceptual jump: θ becomes random

This is the step that trips everyone, so go slowly.

Everywhere in the first half of the course, θ was a **fixed unknown constant** and X was random. That is why we wrote $\mathbb{E}_\theta$ — the subscript said "the distribution depends on this fixed number θ".

To average the risk over θ, we need a weighting function on Θ. Choose one and call it $\pi(\theta)$, and require $\int_\Theta \pi(\theta)\, d\theta = 1$. But a non-negative function integrating to 1 *is a probability density*. So the moment we decide to average the risk, we have — whether we like it or not — treated θ as a random variable.

Common confusion: "But θ isn't random, it's a fixed unknown number!"

Correct, and the Bayesian machinery does not claim otherwise. Two readings, both legitimate:

(i) Purely technical. π is just a set of weights you picked to turn $R(\cdot, T)$ into a number. Nothing philosophical is being asserted. This is the reading that makes minimax work in Lectures 8–9, where the prior is a *device* for proving a theorem about worst-case risk, and nobody believes in it.

(ii) Subjective. π encodes what you believed about θ before seeing data, and the posterior is your updated belief.

For this exam, reading (i) is enough. When a question says "find the Bayes estimate under prior π", it is a computation, not a worldview.

The four objects, and how they fit together

Once θ is random, you have a *joint* distribution of the pair (X, θ), and it can be factored two ways. Learn this table — it is the entire subject.

Name	**Symbol**	**What it is**
Prior	$\pi(\theta)$	Distribution of θ *before* seeing data. You choose it.
Likelihood	$f(x \mid \theta)$	The model. Distribution of X *given* θ — this is the same f you have used all semester.
Marginal	$m(x) = \int_\Theta f(x \mid \theta)\pi(\theta)\, d\theta$	Distribution of X with θ averaged out. Usually you never compute it.
Posterior	$\pi(\theta \mid x)$	Distribution of θ *after* seeing $X = x$. **This is the object you actually want.**

The joint density is

$$\underbrace{f(x \mid \theta)\,\pi(\theta)}_{\text{factor as (data given }\theta)\times\text{(prior)}} \quad = \text{joint}(x,\theta) = \quad \underbrace{\pi(\theta \mid x)\, m(x)}_{\text{factor as (}\theta\text{ given data)}\times\text{(marginal)}}$$

and rearranging gives Bayes' theorem:

$$\boxed{\pi(\theta \mid x) = \frac{f(x \mid \theta)\,\pi(\theta)}{m(x)} = \frac{f(x \mid \theta)\,\pi(\theta)}{\int_\Theta f(x \mid \theta')\pi(\theta')\, d\theta'}}$$

The single most useful practical fact

The denominator $m(x)$ does *not* involve θ. It is just whatever constant makes $\pi(\theta \mid x)$ integrate to 1. So you almost never compute it. Instead write

$$\pi(\theta \mid x) \;\propto\; f(x \mid \theta)\,\pi(\theta) \qquad \text{i.e.} \qquad \textbf{posterior} \;\propto\; \textbf{likelihood} \times \textbf{prior},$$

throw away every factor that does not contain θ, and *recognise the shape* of what is left. That is 90% of the technique.

3 A complete worked example, slowly

Let us do one from scratch with numbers before any theory.

Setup. You toss a bent coin $n = 10$ times and see $x = 7$ heads. So $X \sim \text{Bin}(10, \theta)$ and you want to estimate $\theta = \mathbb{P}(\text{head})$.

Step 1: choose a prior. Say you believe the coin is probably not wildly biased, but you are not confident. The Beta(2, 2) density, $\pi(\theta) \propto \theta(1-\theta)$ on $(0, 1)$, is symmetric about $\frac{1}{2}$ and gently humped — a reasonable "mildly sceptical of extremes" choice. Its mean is $\frac{1}{2}$.

Step 2: write down likelihood × prior, dropping constants.

$$\pi(\theta \mid x = 7) \;\propto\; \underbrace{\binom{10}{7}\theta^7(1-\theta)^3}_{\text{likelihood}} \cdot \underbrace{\frac{\theta^{2-1}(1-\theta)^{2-1}}{B(2,2)}}_{\text{prior}} \;\propto\; \theta^7(1-\theta)^3 \cdot \theta^1(1-\theta)^1 = \theta^8(1-\theta)^4.$$

Note what got thrown away: $\binom{10}{7}$ and $1/B(2, 2)$ are pure numbers, free of θ.

Step 3: recognise the shape. A density proportional to $\theta^{a-1}(1-\theta)^{b-1}$ is Beta(a, b). Here $a - 1 = 8$ and $b - 1 = 4$, so

$$\theta \mid X = 7 \;\sim\; \text{Beta}(9, 5).$$

No integration was needed anywhere.

Step 4: report the Bayes estimate, which (Section 4) is the posterior mean. For Beta(a, b) the mean is $a/(a+b)$:

$$\widehat{\theta}_{\text{Bayes}} = \frac{9}{9+5} = \frac{9}{14} \approx 0.643.$$

What just happened — read this twice

Compare three numbers:

$$\text{prior mean} = 0.500, \qquad \widehat{\theta}_{\text{Bayes}} = 0.643, \qquad \text{MLE} = \frac{7}{10} = 0.700.$$

The Bayes estimate sits *between* the prior mean and the data-only estimate. That is not a coincidence — rearranging,

$$\frac{a+x}{a+b+n} = \underbrace{\frac{n}{a+b+n}}_{=10/14} \cdot \underbrace{\frac{x}{n}}_{\text{MLE}} + \underbrace{\frac{a+b}{a+b+n}}_{=4/14} \cdot \underbrace{\frac{a}{a+b}}_{\text{prior mean}},$$

and indeed $\frac{10}{14}(0.7) + \frac{4}{14}(0.5) = 0.5 + 0.1429 = 0.643$. ✓

A Bayes estimate is a weighted average of the data and the prior, with weights proportional to n and to $a+b$. So $a+b$ behaves like a *prior sample size*: a Beta$(2, 2)$ prior is worth about 4 imaginary observations. As $n \to \infty$ the data weight $\to 1$ and the prior is forgotten. This picture is worth more than any formula.

4 Why the posterior mean? (Bayes risk)

We have not yet justified "take the posterior mean". Here is where it comes from.

Definition: Bayes risk

For a prior π and an estimator T,

$$r(\pi, T) \;:=\; \int_{\Theta} R(\theta, T)\, \pi(\theta)\, d\theta \;=\; \mathbb{E}_{\pi}\big[R(\theta, T)\big].$$

$R(\theta, T)$ **is a function of** θ**;** $r(\pi, T)$ **is a single number.** That is the whole point — numbers can be compared and minimised, functions cannot. An estimator minimising $r(\pi, \cdot)$ is called *the Bayes estimate with respect to* π, written T_{π}.

Theorem (the central result of Lecture 7)

Under squared error loss, the Bayes estimate is the posterior mean:

$$T_{\pi}(x) = \mathbb{E}[\theta \mid X = x].$$

Moreover the minimum value achieved is

$$r(\pi, T_{\pi}) = \mathbb{E}_m\big[\operatorname{Var}(\theta \mid X)\big] \qquad \text{(the average posterior variance).}$$

Proof, with every step explained. Start from the definition and write both integrals out:

$$r(\pi, T) = \int_{\Theta}\int_{\mathcal{X}} \big(T(x) - \theta\big)^2 f(x \mid \theta)\, \pi(\theta)\, dx\, d\theta.$$

Now use the two-way factorisation of the joint density, $f(x \mid \theta)\pi(\theta) = m(x)\,\pi(\theta \mid x)$, and swap the order of integration:

$$r(\pi, T) = \int_{\mathcal{X}} m(x)\ \underbrace{\left[\int_{\Theta} \big(T(x) - \theta\big)^2 \pi(\theta \mid x)\, d\theta\right]}_{\text{call this } Q(x)}\ dx.$$

This rewriting is the whole proof. Look at what it says: the Bayes risk is an average, over x, of the quantity $Q(x)$, with non-negative weights $m(x)$.

So to minimise $r(\pi, T)$ it suffices to **minimise** $Q(x)$ **separately for each** x. And for a *fixed* x, the value $T(x)$ is just a number — call it a. So we need

$$\min_a\ \mathbb{E}\big[(a - \theta)^2 \,\big|\, X = x\big].$$

This is the standard fact that the mean minimises expected squared deviation:

$$\mathbb{E}\big[(a - \theta)^2 \mid x\big] = \big(a - \mathbb{E}[\theta \mid x]\big)^2 + \mathrm{Var}(\theta \mid x),$$

which is minimised at $a = \mathbb{E}[\theta \mid x]$, with minimum value $\mathrm{Var}(\theta \mid x)$. Therefore $T_\pi(x) = \mathbb{E}[\theta \mid X = x]$, and $r(\pi, T_\pi) = \int m(x)\,\mathrm{Var}(\theta \mid x) dx = \mathbb{E}_m[\mathrm{Var}(\theta \mid X)]$. $\square$

Why you should care about that last formula

$r(\pi, T_\pi) = \mathbb{E}_m[\mathrm{Var}(\theta \mid X)]$ is how *every* minimax problem in this course is computed. In the $N(\theta, 1)$ problem with prior $N(0, k)$, the posterior variance is the constant $k/(k + 1)$, so the Bayes risk is $k/(k + 1)$ immediately — no integration. Memorise it.

Common confusion: "Is the Bayes estimate always the posterior mean?"

No — *only under squared error loss.* The loss function decides which feature of the posterior you report:

Loss $L(\theta, a)$	Bayes estimate	Why
$(\theta - a)^2$ (squared error)	posterior **mean**	mean minimises squared deviation
$\lvert\theta - a\rvert$ (absolute error)	posterior **median**	median minimises absolute deviation
$\mathbb{I}\{\lvert\theta - a\rvert > \epsilon\}$ (0–1)	posterior **mode** (MAP)	mode maximises local mass

This course uses squared error throughout, so "posterior mean" is right here — but say *under squared error loss* in your answer. It is a free mark.

5 Conjugacy: making the computation easy

In principle you now know everything. In practice the integral defining $m(x)$ is usually horrible. **Conjugate priors** are the families for which it is free.

Definition

A family Π of priors is **conjugate** for the model $f(x \mid \theta)$ if

$$\pi \in \Pi \quad \Longrightarrow \quad \pi(\cdot \mid x) \in \Pi \quad \text{for every } x.$$

That is: the posterior belongs to the same family as the prior, only with updated parameters. Updating then reduces to arithmetic on the hyperparameters.

How to find or verify a conjugate family — the kernel-matching drill

1. Write the likelihood and **delete every factor free of θ**. What survives is the *kernel*.
2. Ask: what family of θ-densities has this same functional shape? That family is the conjugate one.
3. Multiply prior kernel $\times$ likelihood kernel and collect exponents.
4. Read off the updated parameters. Never integrate.

5.1 Beta–Binomial

$X \sim \text{Bin}(n, \theta)$. Likelihood kernel: $\theta^x(1-\theta)^{n-x}$ — a power of θ times a power of $(1-\theta)$. Which densities on $(0,1)$ look like that? $\text{Beta}(a,b) \propto \theta^{a-1}(1-\theta)^{b-1}$. Multiply:

$$\theta^x(1-\theta)^{n-x} \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{(a+x)-1}(1-\theta)^{(b+n-x)-1}$$

$$\boxed{\theta \mid x \sim \text{Beta}(a+x,\ b+n-x), \qquad T_\pi = \frac{a+X}{a+b+n}}$$

Interpretation: add the successes to a, the failures to b.

5.2 Gamma–Poisson

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poi}(\lambda)$, $S = \sum X_i$. Likelihood kernel: $e^{-n\lambda}\lambda^S$ — a power of λ times $e^{-\text{const}\cdot\lambda}$. Which densities on $(0, \infty)$ look like that? $\text{Gamma}(\alpha, \text{scale } \beta) \propto \lambda^{\alpha-1}e^{-\lambda/\beta}$. Multiply:

$$e^{-n\lambda}\lambda^S \cdot \lambda^{\alpha-1}e^{-\lambda/\beta} = \lambda^{(\alpha+S)-1}\exp\Big\{-\lambda\Big(n+\frac{1}{\beta}\Big)\Big\}$$

$$\boxed{\lambda \mid \mathbf{x} \sim \text{Gamma}\Big(\alpha+S,\ \frac{\beta}{1+n\beta}\Big), \qquad T_\pi = \frac{\beta(\alpha+S)}{1+n\beta}}$$

Numerical check. $n=5$, $S=12$, prior $\text{Gamma}(2,1)$ (mean 2). Posterior is $\text{Gamma}(14, \frac{1}{6})$ with mean $14/6 \approx 2.333$. Compare MLE $= 12/5 = 2.4$ and prior mean $= 2$: again strictly between. ✓

5.3 Normal–Normal

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$ with σ^2 *known*; prior $\theta \sim N(\mu, \tau^2)$. The exponent is a sum of two quadratics in θ, so completing the square gives a normal again. The cleanest way to state the answer uses **precision** $= 1/\text{variance}$:

$$\boxed{\text{posterior precision} = \frac{n}{\sigma^2} + \frac{1}{\tau^2}, \qquad \text{posterior mean} = \frac{\frac{n}{\sigma^2}\overline{X} + \frac{1}{\tau^2}\mu}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}}$$

In words: *precisions add, and the posterior mean is the precision-weighted average of the data mean and the prior mean.* If you remember only one thing about Normal–Normal, remember that sentence — you can reconstruct the algebra from it.

Numerical check. $n=4$, $\overline{X}=10$, $\sigma^2=4$ (so $\sigma^2/n = 1$, data precision 1); prior $N(8,4)$ (precision $\frac{1}{4}$). Posterior precision $= 1 + \frac{1}{4} = 1.25$, so posterior variance $= 0.8$; posterior mean $= \dfrac{1\cdot 10 + \frac{1}{4}\cdot 8}{1.25} = \dfrac{12}{1.25} = 9.6$. Between the prior mean 8 and the data mean 10, closer to the data because the data are four times as precise. ✓

5.4 Summary table

Model	Conjugate prior	Posterior	Bayes estimate
$\mathrm{Bin}(n,\theta)$	$\mathrm{Beta}(a,b)$	$\mathrm{Beta}(a+x,\, b+n-x)$	$\dfrac{a+X}{a+b+n}$
$\mathrm{Poi}(\lambda)$, n obs	$\mathrm{Gamma}(\alpha,\beta)$	$\mathrm{Gamma}\big(\alpha+S,\, \frac{\beta}{1+n\beta}\big)$	$\dfrac{\beta(\alpha+S)}{1+n\beta}$
$N(\theta,\sigma^2)$, n obs	$N(\mu,\tau^2)$	$N\big(\text{prec. wtd. mean},\ (\frac{n}{\sigma^2}+\frac{1}{\tau^2})^{-1}\big)$	precision-weighted mean
$\mathrm{Exp}(\theta)$, n obs	Inverse-Gamma(α,β)	Inv-Gamma$(\alpha+n,\ \beta+S)$	$\dfrac{\beta+S}{\alpha+n-1}$
$U[a,\theta]$, n obs	$\mathrm{Pareto}(\alpha,a)$	$\mathrm{Pareto}\big(\alpha+n, \max(a, X_{(n)})\big)$	$\dfrac{(\alpha+n)M}{\alpha+n-1}$

Common confusion: "Do I have to memorise this table?"

No, and you should not try. Every row is 30 seconds of kernel matching. What you *must* know cold is the *drill*: strip constants, match the shape, add the exponents. In the exam, derive the row you need — it is more reliable than a half-remembered formula, and derivations earn marks that quoted answers do not.

Common confusion: "Why can't the Cauchy have a nice conjugate family?"

Because conjugacy is really a property of *exponential families*. In an exponential family, $\prod_i f(x_i \mid \theta)$ depends on the data only through $\sum_i T(x_i)$ and n, so a two-parameter prior family suffices — and the update is just "add $\sum T(x_i)$ to one hyperparameter and n to the other". Look back at the table: every update has exactly that form.

The Cauchy is not an exponential family, so its likelihood cannot be compressed into a fixed number of summaries. A conjugate family still exists (products of Cauchy kernels), but it has to grow by one hyperparameter per observation — useless in practice. This is the point of the "find a conjugate family for any density" exercise in Lecture 7: the class of *all* densities is trivially conjugate, and equally useless. **Conjugacy is only valuable when the family is small.**

6 Three properties you will be asked about

6.1 A Bayes estimate is essentially never unbiased

You saw it numerically in Section 3: the estimate is pulled toward the prior mean. That pull is *shrinkage*, and shrinkage is exactly bias. Concretely, for Beta–Binomial,

$$\mathbb{E}_\theta\left[\frac{a+X}{a+b+n}\right] = \frac{a+n\theta}{a+b+n},$$

which equals θ for all θ only if $a=0$ and $a+b=0$ — not a proper prior.

The general argument is four lines and is a standard exam question. Suppose T is both Bayes ($T=\mathbb{E}[\theta \mid X]$) and unbiased ($\mathbb{E}[T \mid \theta]=\theta$). Write $\mathbb{E}_m$ for expectation under the joint law and condition two different ways:

$$\mathbb{E}_m[T\theta] = \mathbb{E}\big[\theta\, \mathbb{E}(T \mid \theta)\big] = \mathbb{E}[\theta^2], \qquad \mathbb{E}_m[T\theta] = \mathbb{E}\big[T\, \mathbb{E}(\theta \mid X)\big] = \mathbb{E}[T^2].$$

Then

$$\mathbb{E}_m\big[(T-\theta)^2\big] = \mathbb{E}[T^2] + \mathbb{E}[\theta^2] - 2\mathbb{E}_m[T\theta] = 0,$$

which forces $T=\theta$ almost surely — impossible unless the problem is degenerate.

Common confusion: "So Bayes estimates are worse than UMVUEs?"

No. Bias is not a defect, it is a *trade*. Shrinking toward the prior costs you a little bias and buys you a lot of variance reduction, and MSE is the sum of the two. You already met this in Lecture 1: $\frac{1}{n+1}\sum(X_i - \overline{X})^2$ is biased and beats the unbiased S^2 at every σ^2. And in Lecture 8 the *minimax* estimator of a binomial proportion is a biased Bayes estimate, not the UMVUE. Unbiasedness is a constraint we impose for tractability, not a virtue.

6.2 The posterior depends on the data only through a sufficient statistic

If $f(\mathbf{x} \mid \theta) = g(\theta, T(\mathbf{x}))\, h(\mathbf{x})$ by the factorisation theorem, then

$$\pi(\theta \mid \mathbf{x}) = \frac{g(\theta, T(\mathbf{x}))\, h(\mathbf{x})\, \pi(\theta)}{\int g(\theta', T(\mathbf{x}))\, h(\mathbf{x})\, \pi(\theta')\, d\theta'} = \frac{g(\theta, T(\mathbf{x}))\, \pi(\theta)}{\int g(\theta', T(\mathbf{x}))\, \pi(\theta')\, d\theta'},$$

since $h(\mathbf{x})$ cancels between numerator and denominator. So the posterior — and hence every Bayes estimate — is a function of T alone. Check the table: every entry depends on the data only through $\sum X_i$, $\overline{X}$, or $X_{(n)}$. ✓

6.3 Improper priors and generalized Bayes

Sometimes you want to express "no prior information". The natural choice on $\theta \in \mathbb{R}$ is $\pi(\theta) \equiv 1$ — but $\int_{\mathbb{R}} 1\, d\theta = \infty$, so this is **not** a probability density. Such a π is called *improper*.

Remarkably, the posterior can still be a perfectly good density. If it is, its mean is called the **generalized Bayes estimate**.

The three examples from the course

1. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$, $\pi \equiv 1$. Then $\pi(\theta \mid \mathbf{x}) \propto \exp\{-\frac{n}{2}(\theta - \bar{x})^2\}$, i.e. $N(\bar{x}, 1/n)$ — proper. Generalized Bayes estimate: $\boxed{\overline{X}}$. Note this is also the MLE and the UMVUE. (Sanity check on Normal–Normal: let $\tau^2 \to \infty$, so prior precision $\to 0$ and the posterior mean $\to \overline{X}$. ✓)

2. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} U[\theta - \frac{1}{2}, \theta + \frac{1}{2}]$, $\pi \equiv 1$. The likelihood is the indicator $\mathbb{I}[x_{(n)} - \frac{1}{2} \le \theta \le x_{(1)} + \frac{1}{2}]$, so the posterior is *uniform* on that interval and the estimate is its midpoint: $\boxed{\frac{1}{2}(X_{(1)} + X_{(n)})}$, the midrange. (This is Homework 2 Q3.)

3. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$, $\pi(\theta) = 1/\theta$. Posterior is Inverse-Gamma(n, S) with mean $\boxed{S/(n-1)}$ for $n \ge 2$ — note this is *not* $\overline{X}$.

Common confusion: "If the prior is improper, is the estimate still Bayes?"

Not in the technical sense — and this matters in Lecture 8. The theorem "Bayes with constant risk $\Rightarrow$ minimax" needs π to be a *probability* measure, because its proof uses "sup $\ge$ average". Under an improper prior the Bayes risk is infinite and that step is meaningless. That is exactly why X being minimax for $N(\theta, 1)$ needs the *sequence* argument with proper priors $\pi_k = N(0, k)$, rather than the one-line constant-risk argument. If you are asked "can a generalized Bayes equalizer estimate be minimax?", the answer is: yes it can, but you must prove it via limits of proper Bayes rules, not via the constant-risk theorem.

7 How Bayes connects to the rest of the course

The map

$$\textbf{Bayes} \xrightarrow[\text{take } \sup_\theta]{\text{constant risk}} \textbf{Minimax}$$

The reason Bayes appears immediately before minimax is that *Bayes estimates are the tool for proving minimaxity.* Two theorems:

1. **Constant-risk criterion.** If T_π is Bayes for a proper π and $R(\theta, T_\pi) = c$ for all θ, then T_π is minimax. Proof: $\sup_\theta R(\theta, T) \geq r(\pi, T) \geq r(\pi, T_\pi) = c = \sup_\theta R(\theta, T_\pi)$.
2. **Limit of Bayes rules.** If $r(\pi_k, T_{\pi_k}) \to c$ and $\sup_\theta R(\theta, T^*) = c$, then T^* is minimax.

In both, the middle inequality "sup $\geq$ average" is what does the work — and averaging is precisely what the prior was introduced for in Section 1. The circle closes.

8 Answering an exam question

The standard six lines

1. Write **posterior** $\propto$ **likelihood** $\times$ **prior**.
2. Delete every factor free of θ.
3. Collect exponents; **name** the resulting standard density.
4. Quote its mean. State "*the Bayes estimate under squared error loss is the posterior mean*".
5. If asked about bias, compute $\mathbb{E}_\theta T$ directly and observe it is not θ.
6. If asked for the Bayes *risk*, use $r(\pi, T_\pi) = \mathbb{E}_m[\mathrm{Var}(\theta \mid X)]$ — take the posterior variance and average it over the marginal.

9 Practice, with answers

Do these with the answers covered. They take about 25 minutes together.

P1. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{Bern}(\theta)$ with prior $\mathrm{Beta}(a, b)$. Find the Bayes estimate, express it as a weighted average of the MLE and the prior mean, and say what happens as $n \to \infty$.

P2. $X \sim$ Geometric, $f(x \mid \theta) = \theta(1-\theta)^x$, $x = 0, 1, 2, \ldots$ Find a conjugate family and the Bayes estimate of θ.

P3. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2)$, σ^2 known, prior $N(\mu, \tau^2)$. Show the posterior variance is strictly smaller than both σ^2/n and τ^2. Interpret.

P4. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} U[0, \theta]$ with prior $\mathrm{Pareto}(\alpha, m)$, i.e. $\pi(\theta) = \alpha m^\alpha / \theta^{\alpha+1}$ for $\theta > m$. Find the posterior and the Bayes estimate.

P5. $X \sim N(\theta, 1)$ with prior $N(0, k)$. Find the Bayes estimate and show the Bayes risk equals $k/(k+1)$. What happens as $k \to \infty$, and why does it matter?

Answers

P1. Posterior $\propto \theta^S(1-\theta)^{n-S}\theta^{a-1}(1-\theta)^{b-1}$ with $S=\sum X_i$, so Beta$(a+S,\ b+n-S)$ and

$$T_\pi = \frac{a+S}{a+b+n} = \frac{n}{a+b+n}\cdot\overline{X} + \frac{a+b}{a+b+n}\cdot\frac{a}{a+b}.$$

As $n\to\infty$ the data weight $\to 1$, so $T_\pi - \overline{X}\to 0$: **the prior washes out**. This is why the choice of prior matters little for large samples and a lot for small ones.

P2. Likelihood kernel $\theta(1-\theta)^x$ — a power of θ times a power of $(1-\theta)$, so try Beta(a,b) again:

$$\theta(1-\theta)^x\cdot\theta^{a-1}(1-\theta)^{b-1} = \theta^{(a+1)-1}(1-\theta)^{(b+x)-1},$$

so $\theta \mid x \sim$ Beta$(a+1,\ b+x)$ and $T_\pi = \dfrac{a+1}{a+b+x+1}$. The Beta family is conjugate for the geometric too — as it must be, since both are exponential families in θ with the same $(1-\theta)$ structure.

P3. Posterior variance $=\left(\dfrac{n}{\sigma^2}+\dfrac{1}{\tau^2}\right)^{-1} = \dfrac{\sigma^2\tau^2}{n\tau^2+\sigma^2}$. Then

$$\frac{\sigma^2\tau^2}{n\tau^2+\sigma^2} < \frac{\sigma^2}{n} \iff n\tau^2 < n\tau^2+\sigma^2\ \checkmark, \qquad \frac{\sigma^2\tau^2}{n\tau^2+\sigma^2} < \tau^2 \iff \sigma^2 < n\tau^2+\sigma^2\ \checkmark.$$

Interpretation: precisions add, so combining two sources of information is always at least as good as either alone. The posterior is sharper than the data alone *and* sharper than the prior alone.

P4. Likelihood $\theta^{-n}\mathbb{I}[\theta\ge x_{(n)}]$, prior $\propto \theta^{-(\alpha+1)}\mathbb{I}[\theta>m]$. Multiply:

$$\pi(\theta\mid \mathbf{x}) \propto \theta^{-(\alpha+n)-1}\,\mathbb{I}\big[\theta > \max(m, x_{(n)})\big]$$

which is Pareto$\big(\alpha+n,\ M\big)$ with $M=\max(m, X_{(n)})$. A Pareto(β, M) has mean $\beta M/(\beta-1)$ for $\beta>1$, so

$$T_\pi = \frac{(\alpha+n)\,M}{\alpha+n-1}.$$

Note the shape: like the UMVUE $\frac{n+1}{n}X_{(n)}$, it inflates the largest observation — sensible, since $X_{(n)}$ always *under*estimates θ.

P5. Precision form: data precision 1, prior precision $1/k$, so posterior precision $=1+\frac{1}{k}=\frac{k+1}{k}$ and

$$\theta\mid X=x \ \sim\ N\Big(\frac{k}{k+1}x,\ \frac{k}{k+1}\Big), \qquad T_\pi = \frac{k}{k+1}X.$$

The posterior variance is the *constant* $k/(k+1)$ (it does not involve x), so

$$r(\pi_k, T_{\pi_k}) = \mathbb{E}_m\big[\mathrm{Var}(\theta\mid X)\big] = \frac{k}{k+1}.$$

As $k\to\infty$ this $\to 1$, which is exactly $\sup_\theta R(\theta, X)=1$. By the limit-of-Bayes-rules theorem, X is **minimax**. This is Homework 2 Q1, and it is the single most important place Bayes machinery is used in this course — note that X itself is not Bayes for any proper prior; the sequence is what does the work.

Companions: *`PI_Exam_Revision.pdf` §8–§9, `PI_Class_Exercises_Solved.pdf` §6–§7, `PI_Mock_Exam.pdf` Q5 and Q6.*